

A Comprehensive Analysis Report on the Adoption of Emerging Technologies

DCIT 26 – Application Development and Emerging Technologies

Bachelor of Science in Computer Science

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1. Introduction

Technologies continue to develop and influence society, business, education, and industry. Other technologies are considered emerging because they continue to develop, grow rapidly, and have the potential to create significant change in the future.

Emerging technologies can refer to technologies that are currently developing or may develop within the next five to ten years, and can substantially change the business and social environment. But a technology does not really need to be completely new to be considered emerging.

Organizations should not adopt emerging technologies simply because they are new. Instead, organizations should adopt technology based on business requirements, expected return on investment, security, available resources, regulatory requirements, and the organization's readiness. This report discusses emerging technologies and their importance to business, industries, and society.

2. Characteristics of Emerging Technologies

According to Rotolo, Hicks, and Martin (2015), emerging technologies have five main characteristics. This five-point framework serves specific strategic and analytical purposes for governments, businesses, and researchers.

Characteristics	Description
Radical Novelty	Technology can involve progressing technology or existing technology being used in new ways.
Relatively Fast Growth	Emerging technologies may experience rapid growth in users, research, funding, products, and services.
Coherence	Emerging technologies should involve the combination of previously separate technologies or research areas.
Prominent Impact	They can significantly affect industries, businesses, institutions, and society.
Uncertainty and Ambiguity	Their future applications and effects may not yet be completely known.

Table 1. Characteristics of an Emerging Technology

These characteristics show that emerging technology is not simply about building something new. It also involves continuous development, rapid growth, potential impact, and uncertainty about future applications.

3. Major Emerging Technologies

3.1 Artificial Intelligence

Artificial Intelligence is a technology that lets computers and machines can simulate human comprehension, learning, decision-making, and creativity.

Applications of AI

- Data Analysis
- Transportation
- Customer Service
- Education
- Healthcare
- Research

Artificial intelligence can have a significant effect on different sectors because it can be applied to a wide variety of tasks and problems. It has a great impact on businesses, education, healthcare, and individuals.

3.2 Machine Learning

Machine Learning is a subset of AI that can "learn" from experience and patterns of data. It is used in different fields like healthcare, finance, marketing, and transportation.

One example is in healthcare: machine learning algorithms are used to help identify and develop new medicines more efficiently.

Potential Benefits

- Faster analysis of information
- Automation of certain tasks
- Support for decision-making
- Applications across multiple industries
- Ability to learn from data

3.3 Augmented Reality

Augmented Reality (AR) is a technology that integrates digital information into the user's environment. It combines computer-generated elements with the real world and creates an enhanced view

Examples of AR Applications

- **Navigation and driving:** In-car displays can provide driving and parking guidance.
- **Entertainment:** Games such as Pokémon GO use AR.
- **Shopping:** Customers can see how furniture might look inside their homes before purchasing.
- **Workplace:** AR can provide workers with real-time information and instructions.

AR is an important component of Industry 4.0.

3.4 Quantum Computers

Quantum computing is an advanced machine that uses the rules of quantum mechanics to process information differently. Unlike conventional computers, quantum computers use qubits as their basic unit of computation.

Potential Applications

- Scientific research
- Simulation of atoms and molecules
- Materials design
- Complex calculations
- Development of quantum software

Quantum Computing is still in development; however, powerful quantum computers could potentially contribute to scientific and technological advancements in the future.

4. Emerging Technologies in Industry 4.0

The Fourth Industrial Revolution (Industry 4.0) is the developing and current stage of industrial technology. It involves disruptive technologies such as IoT, VR, AR, and AI, which are changing the way people live and work.

Technology	Purpose/Application
Big Data Analytics	Collects, organizes, and analyzes large amounts of data to identify patterns and support decision-making.

Internet of Things (IoT)	Connects physical objects to the internet so they can communicate and interact with other devices.
Cloud Computing	Provides hosted computing services through the internet.
Augmented Reality	Provides workers with real-time information and instructions through digital overlays.

Table 2. Emerging Technologies in Industry 4.0

These technologies can help organizations improve production, decision-making, and customer management.

5. Conclusion

The emergence of new technologies is an ongoing process whereby these technologies evolve and grow over time. Therefore, they can transform the way we live and work. The primary features of an emerging technology are as follows: (Radical innovation – it implements something that is altogether novel or dramatically divergent from what existed before; Rapidly increasing adoption – such technologies are therefore progressively developing rapidly and becoming generally accepted only within a few years; Coherence – emerging technology is usually part of a broader category and may be-motivated by earlier technological innovations; Significant impact – emerging technology would greatly change the way of business, organization, and societies over the long-term; Uncertainty – the precise ways in which emerging technology would shape the world in the next few decades are still unknown at this point.

Examples of the above-mentioned types of emerging technology that illustrate how modern technology is constantly growing and improving in the present time are Artificial Intelligence, Machine Learning, Computer-Based Augmented Reality, Quantum Computing, Big Data Analytics, Internet of Things (IoT), and Cloud Computing. They not only represent some of the core technological systems that would push Industry 4.0 toward a future direction but also play an integral role and must be integrated within the process for companies to develop.

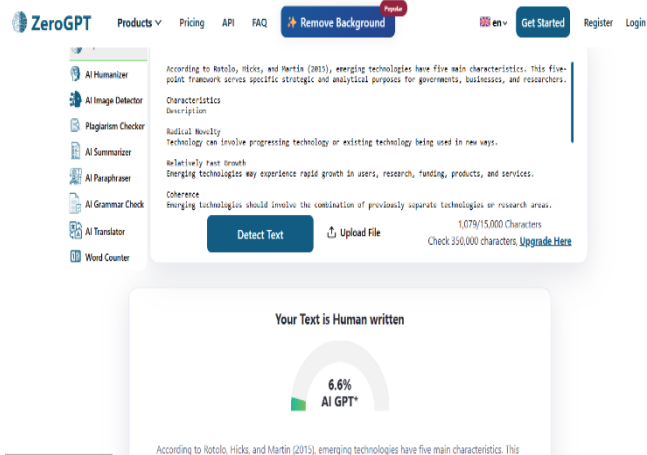
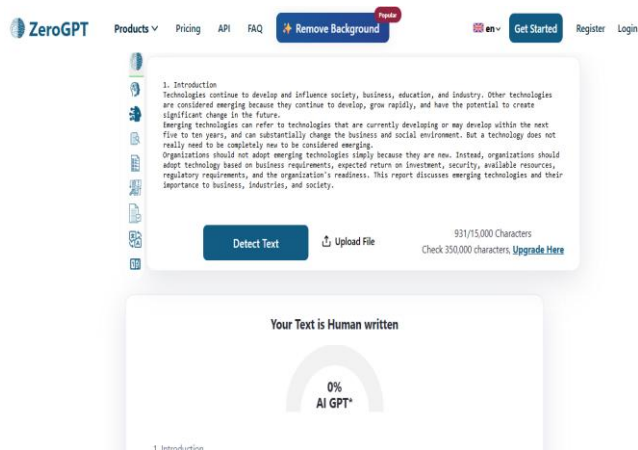
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